

Riverdale Public Utility District

Revised Groundwater Monitoring Well Installation Workplan

Fresno County, CA
October 2018

Prepared for:
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Prepared by:
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Abbreviations

bgs	below ground surface
DWR.....	Department of Water Resources
MCL.....	maximum contaminant level
mg/L.....	milligrams per liter
MRP	Monitoring and Reporting Program
NO ₃ -N	nitrate as nitrogen
PUD	Public Utility District
RWQCB.....	Regional Water Quality Control Board
TDS.....	total dissolved solids
WDR.....	Waste Discharge Requirements
WWTP.....	Wastewater Treatment Plant

1 Introduction

The following Revised Groundwater Monitoring Well Installation Workplan (Revised Workplan) has been prepared by Provost & Pritchard Consulting Group (Provost & Pritchard) for the Riverdale Public Utility District (PUD) Wastewater Treatment Plant (WWTP or Site) located east of the Brawley Avenue alignment, and south of Cerini Avenue approximately one mile north of the Riverdale community in Fresno County, California. The Site is located in Section 13, Township 17 South, Range 19 East, Mount Diablo Baseline and Meridian. See **Appendix A: Figure 1. Vicinity Map**.

A previous Groundwater Monitoring Workplan (Workplan) was prepared by Provost & Pritchard in December 2006 and approved by the Regional Water Quality Control Board (RWQCB) in correspondence dated 14 February 2007. Due to the necessity to hold a public bidding process in which no bids were received, and subsequent interruptions to facility upgrades, the 2006 Workplan was not implemented. The RWQCB indicated that well installation could be delayed and installed with WWTP Upgrades. This Revised Workplan is to address changes in the conditions underlying the Site which have occurred in the twelve years interim, and to reflect updated procedures to be implemented at the Site.

This Revised Workplan has been prepared under the direct supervision of a Certified Hydrogeologist registered in the State of California in accordance with the RWQCB Standard Requirements for Monitoring Well Installation Reports.

2 Groundwater Conditions Summary

2.1 Purpose

The purpose of the well installation project is to establish a monitoring well network for general compliance with Waste Discharge Requirements (WDRs) Order No. R5-2013-0096 and its included Monitoring and Reporting Program (MRP).

2.2 Geologic and Hydrogeologic Conditions

The Site is located within the Tulare Lake Subbasin of the San Joaquin Valley Groundwater Basin. The San Joaquin Valley is a structural trough, approximately 70 miles in width by 200 miles in length, filled with up to 32,000 feet of marine and continental sediments deposited during periodic inundation by the Pacific Ocean and erosion of the surrounding mountains. The site overlies a portion of the aquifer that has a total thickness of about 2,000 feet, based on the depth to the top of saline water as mapped by Page (USGS, 1973), and the most current depth to water contours available from the Department of Water Resources (DWR) for spring 2017.

The San Joaquin Valley is bounded by the Coast Ranges to the west, the San Emigdio and Tehachapi Mountains to the south, the Sierra Nevada to the east, and the Sacramento-San Joaquin Delta and Sacramento Valley to the north. The northern San Joaquin Valley drains towards the Delta by way of the San Joaquin, Fresno, Merced, Tuolumne, and Stanislaus Rivers. The southern San Joaquin Valley drains internally by way of the Kings, Kaweah, Tule, and Kern Rivers into the Tulare drainage basin, which includes the beds of the former Tulare, Buena Vista, and Kern Lakes.

The sediments that comprise the aquifer system consist of Tertiary and Quaternary age unconsolidated continental deposits. The Quaternary deposits are generally divided into older alluvium and lacustrine deposits, younger alluvium, and flood-basin deposits. The older alluvium consists of interbedded lenses of clay, silt, silty/sandy clay, clayey/silty sand, sand, gravel, cobbles, and boulders. The younger alluvium consists of fluvial arkosic beds interbedded with flood-basin deposits (DWR, 2006).

Clay beds of the lacustrine and marsh deposits form aquitards that control the vertical and lateral movement of groundwater. The most prominent clay bed is the Corcoran clay, or E-clay, which was formed as a characteristically dark greenish, blue gray silty clay that is stiff and nearly impermeable. The thickness of the Corcoran Clay ranges from about 10 feet near its edge to more than 160 feet beneath the Tulare Lake bed (USGS, 1986). In this area the aquifer is commonly hydrogeologically divided into a confined to semi-confined aquifer below the Corcoran clay and an unconfined aquifer above the Corcoran clay. The depth to the top of the Corcoran Clay beneath the site is between 435 and 475 feet below ground surface (bgs). The site is outside of the area underlain by the A clay, a regional near surface clay layer, as mapped by Croft (USGS, 1972).

Groundwater level conditions were evaluated from DWR *Lines of Equal Elevation of Water in Wells, Unconfined Aquifer, San Joaquin Valley* groundwater elevation data, which are summarized in Table 2.1 and presented as groundwater contour maps in **Appendix B**. Since 2011 the groundwater flow direction has been dominantly westerly to northwesterly. The most recent groundwater level data is for Spring 2018, at which time the site fell between the 210-220 feet bgs contours.

Table 2.1 Groundwater Flow Direction, Elevation, and Depth to Water

Year	Season	Flow Direction	Groundwater Elevation (ft)	Depth to Water (ft)
2010	Spring	W	50-60	170
2011	Spring	W	60-70	160
2012	Spring	W	70	150-160
2013	Spring	NW	60	150-160
2013	Fall	W	50	170
2014	Spring	W-NW	50-60	170
2014	Fall	W	30	190
2015	Spring	W	50	180
2015	Fall	W	30	190
2016	Spring	W	0-10	220
2016	Fall	W-SW	10	210
2017	Spring	NW	40	180
2017	Fall	W-NW	40	180
2018	Spring	NW	0-10	210-220
Note: Groundwater elevation based on nearest groundwater contour if one does not traverse the site.				

2.3 Groundwater Quality

An Antidegradation Analysis completed by Provost & Pritchard in 2017 indicates that nitrate as nitrogen (NO₃-N) is generally very low upgradient of the site (0 to 3.1 milligrams per liter [mg/L]), while the highest concentrations were reported from a well about 5 miles north to northwest of the Site at 15.6 mg/L. The same report used three wells from the vicinity of Riverdale to characterize the groundwater quality for the area, the results of which are displayed in Table 2.2 (P&P, 2017). The NO₃-N results were below the maximum contaminant level (MCL) of 10 mg/L, ranging from 0.0 to 0.15 mg/L. Total dissolved solids (TDS) concentrations were consistent with the pre-1969 average of 400 mg/L, with results ranging from 340 to 397 mg/L.

Table 2.2. Current Groundwater Quality Data

				Well 4	Well 5	TW 6 Zone 8	Average
Constituent	Units	Maximum Contaminant Levels	Secondary Standards	Perforated 370' to 490'	Perforated 190' to 500'	300' to 430'	
Conductivity @ 25C	umhos/cm	-	900-1,600	644	621	510	592
Nitrate as NO ₃	mg/L	45	-	0.57	0.67	ND	0.6
Nitrate as NO ₃ -N (calculated)	mg/L	10	-	0.13	0.15	0.0	0.09
TDS	mg/L	-	500-1,000	397	378	340	372
Arsenic	ug/L	10	-	36	37	30	35

Section Two: Groundwater Conditions Summary

Riverdale Public Utility District Revised Groundwater Monitoring Well Installation Workplan

Wells were sampled in the vicinity of the Site for the 2006 Workplan. Reported nitrogen (NO_3) concentrations for this sampling event were 18.6 mg/L, 4.2 mg/L, and <2 mg/L, well below the (MCL) of 45 mg/L NO_3 . The arrangement of these wells indicates that NO_3 concentrations increased northward across the Site. It is possible that the elevated NO_3 concentrations downgradient of the site were impacted by the Zacky Farms chicken ranch located north of the Site.

3 Monitoring Well Installation

Groundwater levels in the San Joaquin Valley have declined dramatically in the last few years raising concern that any monitoring wells installed at this time will need to accommodate widely fluctuating groundwater levels. As such, extended well screens with a minimum length of 80 feet are proposed in order to extend the useful lifespan of the proposed wells should groundwater levels continue to fluctuate to the extent seen in the last several years. Final well design will be based on soil boring lithology and depth to first encountered groundwater.

The monitoring wells will be constructed in general accordance with California Well Standards and local regulations. The proposed groundwater monitoring wells will be constructed of 2-inch schedule 80 PVC with 0.020 machine-slotted well screens. For the purposes of this work plan, 80-foot well screens are proposed. It is anticipated that groundwater will be encountered at approximately 215 feet bgs. The well screens will extend approximately 40 feet above first encountered groundwater and 40 feet below.

A filter pack consisting of Lonestar #3 sand or equivalent will be emplaced from the bottom of the boring up to a minimum of 2 feet above the top of the screened interval. A minimum three-foot thick transition seal will then be emplaced above the filter pack. The transition seal will consist of bentonite chips or pellets hydrated with clean water. An annular seal consisting of a cement grout mixture will then be placed above the bentonite transition seal with the use of a tremie pipe.

Once installed, the well casing will be cut to the required height and capped with a water-tight well cap. A concrete base will be installed at the well head with a water-tight painted steel standpipe and a locking well cover as shown in **Appendix A: Figure 3**. Well pads will be constructed so that surface drainage will slope away from the casing.

The monitoring well installation will be performed in accordance with the Monitoring Well Installation protocol in **Appendix C**. A construction schematic is presented in **Appendix A: Figure 3**.

3.1 Proposed Monitoring Well Locations and Rationale

One (1) upgradient and two (2) downgradient groundwater monitoring wells are proposed. The proposed locations, as summarized below, are shown on **Appendix A: Figure 2**. Proposed well locations may need to move slightly due to field conditions.

3.1.1 Upgradient Monitoring Locations

Groundwater monitoring well MW-1 is proposed to be installed generally southeast of the Site, at the southeast corner of the existing WWTP. Based on the groundwater flow direction at the Site, MW-1 will provide groundwater quality data upgradient of the land application areas associated with the Site.

3.1.2 Downgradient Monitoring Locations

Downgradient monitoring wells are proposed to be installed along the western and northern boundaries of the land application areas. The proposed monitoring locations are as follows:

- Proposed monitoring well MW-2 is located on the western boundary of the WWTP additional land.

- Proposed monitoring well MW-3 is located on the northern boundary along the facilities' proposed disposal area.

3.2 Drilling Details

Permits will be obtained from Fresno County per the Fresno County Water Well Permitting Program website:

<https://www.co.fresno.ca.us/departments/public-health/environmental-health/water-surveillance-program/water-well-permitting-program>

Site activities will be overseen by a Provost & Pritchard geologist or hydrogeologist registered in the State of California. Well drilling will be performed in general accordance with DWR California Well Standards Bulletins 74-81 and 74-90 and Fresno County requirements.

Specifically, drilling will be in accordance with the Drilling and Soil Sampling protocol in **Appendix C**. Monitoring well installation and development will be performed in accordance with the Monitoring Well Installation, Development, and Sampling protocols in **Appendix C**.

It is anticipated that the wells will be drilled by a mud/air rotary or sonic drilling rig and crew capable of reaching the target depth. For a rotary rig, composite soil sampling to determine lithology will be conducted at 5 to 10-foot intervals or major changes in lithology for the entire length of the soil boring or until saturated conditions make sampling too difficult. For a sonic rig, continuous sampling will be utilized. Geophysical logging may be utilized in the boring to better clarify soil lithology. For the purposes of this Workplan, it is anticipated that groundwater will be encountered at approximately 215 feet bsg.

A project-specific Health and Safety Plan is included in **Appendix D**.

3.3 Well Development

See **Appendix C**: Monitoring Well Installation, Development, and Sampling protocol.

3.4 Well Survey

The survey will be performed by a Provost & Pritchard licensed land surveyor registered in the State of California. A combination of Global Positioning System (GPS) and ground survey methods will be used based on the following datums:

- Horizontal – North American Datum of 1983 (NAD83), California Coordinate System of 1983 (CCS83) state plane coordinates.
- Vertical – North American Vertical Datum of 1988 (NAVD88)

Survey measurements will be reported to +/- 0.01 feet.

3.5 Sampling and Analysis Plan

Once installed and developed, the proposed groundwater monitoring wells will be sampled in accordance with the MRP for the Site. Scheduling and sampling of the monitoring wells will be performed by a

groundwater monitoring subcontractor selected by Riverdale. The methods used for groundwater sampling shall be consistent with the Groundwater Sampling Protocol in **Appendix C**.

3.6 Monitoring Well Installation Completion Report

Provost & Pritchard will prepare and submit a post construction report summarizing the monitoring well installation activities. The report will include as-built monitoring well construction details, boring logs, e-logs (if performed), depth to groundwater data, field notes, a construction summary table, well development procedures and data, well survey data, an updated site plan.

3.7 Schedule for Completion of Work

The monitoring wells will be installed during construction of the WWTP improvement project, after RWQCB approval of this Workplan. Construction of the WWTP is anticipate to begin during the summer of 2019. Based on the anticipated well depths, well installation activities will likely take approximately one to two weeks to complete once drilling begins. Development would follow as soon as the annular seals have set sufficiently as not to be damaged by the development procedures. Following well installation, a post construction report will be submitted within approximately six weeks summarizing the monitoring well installation activities.

4 Limitations

This Workplan has been prepared for the sole use of the client and involved regulatory agencies. Any other person or entity without the express written consent of Provost & Pritchard Consulting Group may not rely upon this report.

Provost & Pritchard's professional services were performed consistent with generally accepted environmental principles and practices in California at the time the services were performed. No guarantee or warranty, expressed or implied, is made.

5 References

California State Water Resources Control Board, Geotracker GAMA website

http://www.swrcb.ca.gov/gama/geotracker_gama.shtml

California Department of Water Resources, *Lines of Equal Elevation of Water in Wells, Unconfined Aquifer, San Joaquin Valley*, for years 2010 through 2018

California Department of Water Resources, San Joaquin District, 1981. *Depth to the Top of the Corcoran Clay (Map) – San Joaquin Valley*.

California Department of Water Resources, 2006. *San Joaquin Valley Groundwater Basin, Kings Subbasin, California's Groundwater Bulletin 118*.

Provost & Pritchard Consulting Group, 2017. *Riverdale Public Utility District Antidegradation Analysis Wastewater Treatment Plant Improvement Project*.

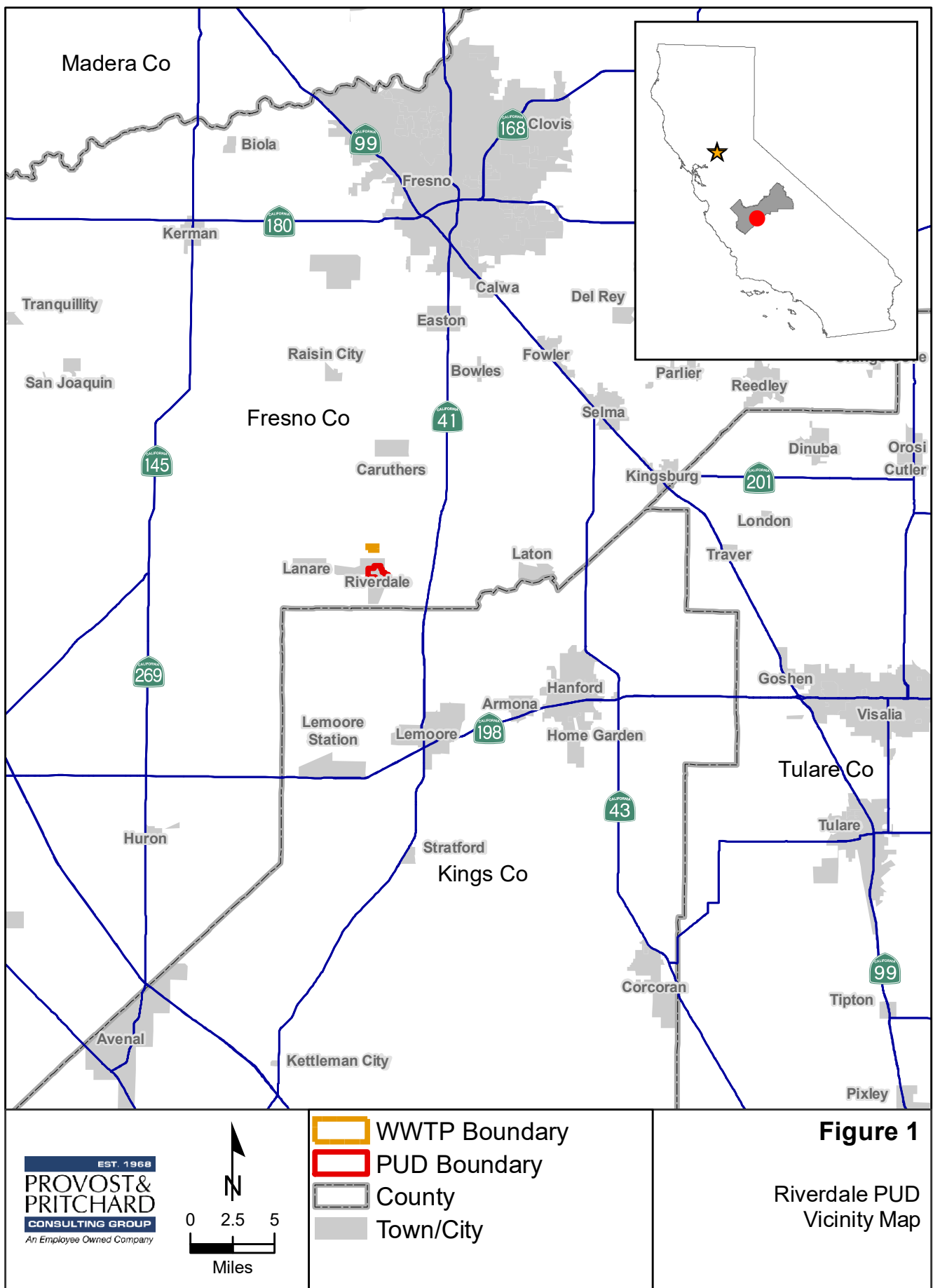
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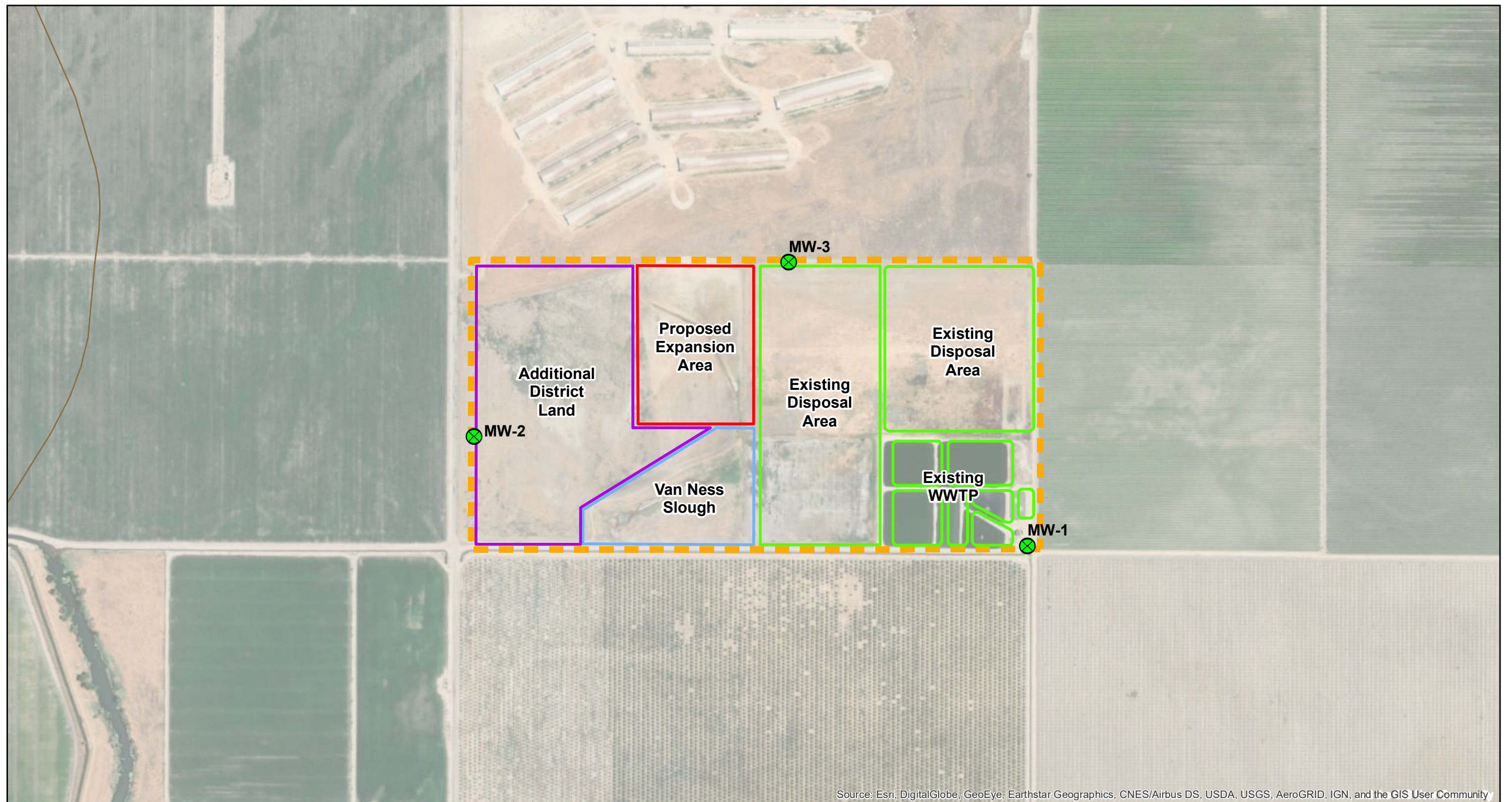
United States Geological Survey in cooperation with the California Department of Water Resources, 1973, Base of Fresh Groundwater (approximately 3,000 micromhos) in the San Joaquin Valley, California, Hydrologic Investigations Atlas HA-489, Page, R.W.

United States Geological Survey, 1986, Geology of the Fresh Ground-Water Basin of the Central Valley, California, with Texture Maps and Sections, Professional Paper 1401-C, Page, R. W.

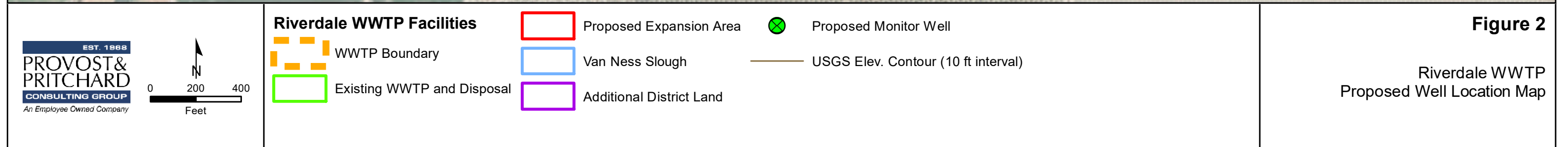
Appendices

Appendix A-Figures



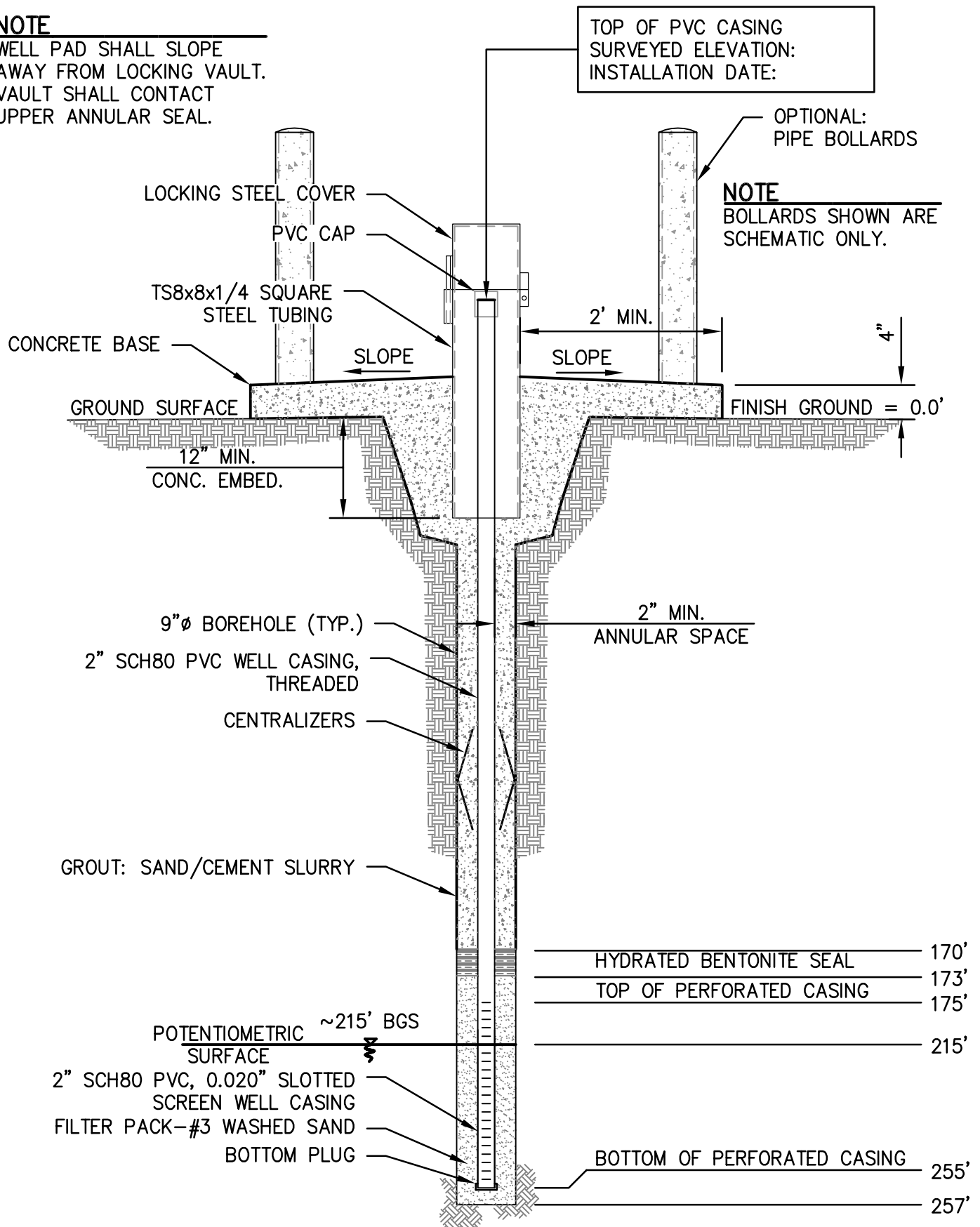


Source: Esri, DigitalGlobe, GeoEye, Earthstar Geographics, CNES/Airbus DS, USDA, USGS, AeroGRID, IGN, and the GIS User Community



NOTE

WELL PAD SHALL SLOPE AWAY FROM LOCKING VAULT. VAULT SHALL CONTACT UPPER ANNULAR SEAL.



TOTAL DEPTH: 257'

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REVISED GROUNDWATER MONITORING
WELL INSTALLATION WORKPLAN
RIVERDALE PUBLIC UTILITY DISTRICT
FRESNO COUNTY

SCHEMATIC MONITORING WELL
PROPOSED DESIGN
FIGURE 3 (MW1-MW2-MW3)

DESIGN ENGINEER:
LGS
DRAFTER: DJD
SCALE: N. T. S.
DATE: 9/14/2018
JOB NO: 2211-16-001
SHEET 1 OF 1

Appendix B- Hydrogeologic Information

See Separate Attachment

Appendix C-Protocols

PROTOCOL FOR DRILLING AND SOIL SAMPLING

GENERAL SCOPE

The procedures used by Provost and Pritchard Consulting Group (Provost & Pritchard) were developed to obtain representative, consistent, reliable, and reproducible data during the drilling, sampling, and sealing of boreholes. The procedures described herein are intended to be of general use. As work progresses, appropriate revisions may be made and approved in writing by Provost & Pritchard's project manager.

PROCEDURES FOR DRILLING

Provost & Pritchard will prepare a work plan for the oversight agency review. Field work and reporting will be performed under the direction of an appropriately registered or certified professional (Business and Professions Code Sections 6735 and 7835). All field personnel have current Hazwoper certification, as applicable. All field equipment will be calibrated prior to field use, and recalibrated, as needed.

Procedures for boreholes intended for conversion into monitoring wells will additionally follow the Provost & Pritchard Monitoring Well Installation, Development, and Sampling Protocol.

1. PERMITTING

Provost & Pritchard will prepare applications for appropriate permits, unless otherwise instructed by the client. Copies of signed permits will be on site during drilling operations, and will be available for inspection by appropriate agencies. The Contractor shall obtain well drilling permits and USA clearance as needed.

2. EQUIPMENT CLEANING

Drilling equipment will be thoroughly steam-cleaned prior to arriving on site to prevent the introduction of contamination from off site. Clean equipment will be stored in a clean location when not in use. Hydrocarbon based lubricants will not be used on drilling equipment. Provost & Pritchard recommends that food grade-solid, vegetable shortening, or a Teflon™ based lubricant be used. Equipment will be steam-cleaned on site, and will be cleaned only in designated areas, prior to reuse.

3. SAMPLER CLEANING

Soil samplers will be disassembled, washed with a solution containing Simple Green or non-phosphate detergent, or steam-cleaned, rinsed with steam distilled water, and air-dried immediately prior to use. Samples will be lined with similarly cleaned and dried brass or stainless steel tubes, and reassembled for use.

4. DRILLING

The drilling method used will be determined based upon the material to be drilled, depth and diameter of the boring, sampling requirements, site access, and other site-specific conditions.

Most wells installed up to approximately 150 feet deep will utilize a hollow-stem auger truck-mounted drilling rig; most deeper wells will require using a mud rotary drilling rig or similar.

The subsurface stratigraphy will be interpreted by observing the materials recovered during drilling, and by sampling undisturbed soils. For hollow-stem drilling, soils will generally be sampled at a minimum of once per each 5 feet drilled, or at noticeable changes in lithology, until ground water is encountered. Soils may be sampled using a California-modified sampler or a standard penetration sampler containing cleaned brass liners. For mud rotary-type drilling, soil cuttings will generally be collected at a minimum of once per each 10 feet drilled to the total depth explored. Geo-physical logs may be performed and utilized for stratigraphic interpretation. Sampling intervals may vary depending upon project requirements. Boring depths will be based upon the nature and extent of the materials encountered, and upon project requirements.

5. PREPARATION OF BORING LOGS

The Provost & Pritchard field technician will lithologically log the borings during drilling. The technician and the drilling operator will discuss changes in material penetrated by the drill, changes in drilling conditions, hydraulic pressure, drilling action, and drilling fluid circulation rate, and the technician will record such changes by time and depth. The technician will evaluate the relative moisture content of the samples and note zones that produce water.

6. SEALING OF BORING

Boreholes will be destroyed by methods as approved by the local oversight agency.

PROCEDURES FOR SOIL SAMPLING

1. EQUIPMENT LIST

The geologist or soil technician will require the following equipment for hollow-stem drilling and soil sampling. Mud rotary samples are generally collected by the drilling subcontractor. In this case, protective equipment and site documents would generally be sufficient.

Decontamination Equipment

- 2 – 5-gallon buckets (or suitable wash tub)
- Simple Green or non-phosphate detergent
- small head long handled scrub brush
- 1 ½" or 2" bottle brush

Drilling Equipment

- chain vise (provided by driller)
- 2 – 5-gallon buckets
- pipe wrench
- rock hammer
- large flathead screwdriver and/or small putty knife
- quart self-sealing plastic bags

- photoionization detector (with a lamp of 11.7 eV), as appropriate
- leather gloves
- eye and ear protection
- clipboard or forms box with the following:
 - client contact information
 - site map with boring locations
 - field boring logs
 - daily field records
 - waterproof fine point marker and ball point pens
 - sample labels and chain-of-custody forms
 - Site Health and Safety Plan

Sampling Equipment

- 1 ½" or 2" stainless steel or brass liners, caps and Teflon tape
- gallon self-sealing plastic bags
- ice chest with ice
- latex or similar gloves
- soil sample collection log (or note on boring log)

2. SAMPLING INTERVAL IN BORINGS

Soil samples may be collected at discrete intervals or by continuous core, based upon project-specific requirements. For hollow-stem drilling with discrete sampling intervals, soils will be sampled utilizing a standard split spoon (SPT), modified California, or California sampler. The sampler will be driven either 18 or 24 inches, as appropriate, ahead of the bottom auger using a slide hammer and rods or other comparable method. The number of blows necessary to drive the sampler each 6 inches will be recorded to help evaluate soil consistency. For continuous core sampling, soils will be sampled by utilizing a continuous split spoon sampler typically either 5 or 10 feet in length. The auger string, with the sampler inside, will be advanced into the soil column a distance equal to the length of the sampler being used.

For both discrete and continuous sampling, once the split spoon has been driven into the soil the appropriate distance, the sampler is retrieved to the surface where it is opened for sample acquisition. Each style of split spoon sampler is designed to contain a set of brass or stainless steel liners appropriate to the length of the sampler. If liners are utilized, soils collected in the bottom liner may be labeled, sealed, and preserved in an ice-cooled container for potential laboratory analyses. Selected soil samples may be delivered under a chain-of-custody record to a State-certified laboratory for chemical or soil property analyses. Soils which are collected in the upper and middle liners may be extruded in the field, and examined by Provost & Pritchard's geologist to help provide detailed lithologic information.

For mud rotary drilling and soil sampling, the drilling subcontractor will collect soil samples for lithologic purposes from the cuttings utilizing a screened scoop or other comparable method. The retained soil will be placed into a self-sealing plastic bag labeled with the drilling depth where the sample was collected.

3. QUALITATIVE FIELD SCREENING

A photoionization detector (PID) may be used to provide a qualitative screening of each soil sample collected during drilling. The field screening procedure typically consists of sealing a soil sample in a self-sealing plastic bag or clean glass jar, allowing the sample to equilibrate for 5 to 15 minutes, and scanning the headspace in the bag or jar for vapors. The screening readings will be noted on the boring log and/or on the daily field log.

4. SAMPLE PRESERVATION

The sample(s) selected for chemical or physical testing will be sealed in the liner in the field. Teflon™ sheeting covers will be placed on the ends of the liner directly on the exposed soils, and held in place by clean plastic caps. Sealed soil samples will be labeled and placed in a covered ice-cooled container.

5. SAMPLE PACKAGING AND TRANSPORT

Sample Handling

Sample containers will not be opened, except by laboratory personnel who will perform the chemical analyses. Soil samples will be analyzed by a laboratory certified by the State of California. Requests for sample analyses will be made in writing and will be included as part of the chain-of-custody record.

Custody Seal

If it is necessary for samples or sample chests to leave the immediate control of the sampler prior to being delivered to the laboratory, a custody seal will be placed on each sample container and/or sample chest to discourage tampering during transportation. The custody seal will contain the sampler's signature, and the date and time seal was emplaced.

Chain of Custody Procedures

In order to document and trace sample possession from time of collection to time of analysis, a chain-of-custody record will be filled out by the sampler, and will accompany the sample through the laboratory analyses. The completed chain-of-custody record will accompany the final laboratory analytical report.

Information contained on the duplicate, carbonless chain-of-custody form will include:

- date and time the sample was collected;
- sample number and the number of sample containers;
- analyses required;
- remarks, including preservatives added and any special conditions; and
- container number in which sample has been packaged.

Blank space on the chain-of-custody record between last sample number listed and signatures at the bottom of the sheet will be lined out.

PROTOCOL FOR MONITORING WELL INSTALLATION, DEVELOPMENT, AND SAMPLING

GENERAL SCOPE

The procedures used by Provost and Pritchard Consulting Group (Provost & Pritchard) for drilling, construction, development, and sampling of groundwater monitoring wells were developed to obtain representative, consistent, reliable, and reproducible data. The wells will generally be drilled by the hollow stem auger method to approximately 150 feet below ground surface, and by mud rotary or similar for deeper wells; other drilling methods may be pre-approved by the Geologist or Engineer. The exact depth and screening interval of each monitoring well will be based on the field conditions encountered and the review and evaluation of past data collected from wells in the vicinity of the investigation.

Upon construction completion, each monitoring well shall be developed and sampled as described in this protocol. The procedures described herein are intended to be of general use. As work progresses, appropriate revisions may be made and approved in writing by Provost & Pritchard's project manager.

PROCEDURES

Field work and reporting will be performed under the direction of an appropriately registered or certified professional (Business and Professions Code Sections 6735 and 7835). All field personnel have current Hazwoper certification, as applicable. All field equipment will be calibrated prior to field use, and recalibrated, as needed.

Provost & Pritchard will prepare a work plan for the oversight agency review, and obtain appropriate permits, as applicable. Subsurface site assessment permits will be completed sufficiently early in the course of the project to allow time for review and authorization at least 48 hours before field activities begin. Copies of signed permits will be on site during sampling activities, and will be available for inspection by appropriate agencies, if applicable.

Complete drilling and sampling procedures are detailed in the Provost & Pritchard Drilling and Soil Sampling Protocol.

1. EQUIPMENT LIST

The geologist or soil technician will require equipment as listed in the Provost & Pritchard Drilling and Soil Sampling Protocol. The water technician will require equipment as listed in the Provost & Pritchard Groundwater Sampling Protocol.

2. QUALIFICATIONS OF CONTRACTORS

The wells shall be drilled by qualified well drillers with experience in construction of monitoring wells. The well driller must possess a current C-57 Well Drillers License, valid in the State of California.

3. STANDARDS AND PERMITS

The wells shall be constructed in accordance with the local County ordinance, and the California Water Well Standards (DWR Bulletins 74-90 and 74-81). The Contractor shall obtain well drilling permits and USA clearance as needed, and shall report the results of the drilling to the California Department of Water Resources.

4. EQUIPMENT REQUIREMENTS

Drilling equipment and soil samplers will be thoroughly washed with a solution containing a non-phosphate detergent, or steam-cleaned, rinsed with steam distilled water, and air-dried immediately prior to use on site to prevent the introduction of contamination from another source. If equipment needs to be steam-cleaned on site, it will be cleaned in a designated area prior to reuse. Soil samplers will be lined with similarly cleaned and dried brass or stainless steel tubes as needed, and reassembled for use.

5. WELL DRILLING

During the groundwater monitoring well drilling process the following information will be recorded and logged:

- A. A detailed log of unique soil horizons and hydrogeological conditions encountered, from the surface to the total depth, as interpreted from soil samples. Physical descriptions of soils excavated will be noted as per the Unified Soils Classification System (USCS).
- B. Discrete soil samples will be collected at five foot intervals from the surface to the total depth, as per standard geological practice, placed in separate bags for each interval, and plainly marked with the well name and depth. Additional discrete or continuous samples may be collected at significant changes in subsurface conditions. Selected soil samples may be delivered under a proper chain-of-custody record to a State-certified laboratory for chemical analyses.
- C. For mud rotary borings, geo-physical logs may be performed to better clarify the soil lithology and determine depth to groundwater.
- D. The final drillers log shall show diameter, wall thickness, depth, and length of casing installed; perforated intervals; details of cemented sections; and other pertinent details.
- E. Permanent records of drilling shall be made by the Contractor in accordance with the laws of the State of California governing the drilling of monitoring wells.

6. WELL CONSTRUCTION MATERIALS

Blank and Perforated Well Casing

The blank casing is typically 2-inch diameter schedule 40 PVC pipe, with perforated casing of 2-inch diameter, schedule 40 PVC pipe of with machined slots of 0.020-inch width. The Geologist or Engineer may propose wells of smaller or larger diameter and schedule based on specific site conditions.

Filter Pack

Filter pack grain size will be appropriate to the selected screen slot width. Filter pack will consist of Lonestar #3 sand or equivalent with less than 2 percent, by weight, passing through a No. 200 sieve for the 0.020-inch width screen. The filter pack will be selected to retain a high percentage of the saturated zone (native materials) to be monitored and will be graded similarly. The filter pack will extend from the bottom of the borehole, along the length of the perforated section, to approximately two feet above the top of the screened interval. The Geologist or Engineer may propose filter pack of smaller or larger grain size based on specific site conditions.

Transition Seal

The transition seal will be a minimum of three feet thick of bentonite chips or pellets. The Geologist or Engineer may propose a different transition seal thickness based on specific site conditions.

Cement Grout

Cement grout will be a mixture of not more than six gallons of clean water per cubic foot of Type II Portland cement, with up to 5% bentonite, to be approved by the Geologist or Engineer.

7. WELL INSTALLATION

Well Casing

The exact monitoring well construction will be determined by the Geologist or Engineer contingent upon the field conditions encountered. The casing shall be installed in such a manner as to avoid damage to the casing and to allow the proper placement of the filter pack and annular seal.

Filter Pack

The filter pack shall be installed opposite the perforated interval by using a tremie pipe or equivalent method. The tremie pipe will initially be lowered to near the bottom of the interval to be filter packed and then slowly pulled up as sand packing proceeds to the desired depth above the perforated interval. The filter pack for the well screen will be surged and additional filter pack material added as needed.

Transition Seal

The transition seal will be installed above the filter pack. Bentonite installed above standing water will be hydrated by the addition of clean water.

Annular Seal

The cement grout mixture will be placed above the bentonite seal. The annular seal from the top of the transition seal to the land surface shall be installed by pumping the cement under pressure through a tremie pipe. During the operation, the tremie pipe will be raised as the annular space is filled with cement. An upper interval may be emplaced by free-fall method as

approved by the local oversight agency. Once the cement has been installed, the well will not be disturbed for at least 48 hours.

Well Head and Pad

The well casing will be cut to the required height and capped with a water resistant well cap. A concrete base will be installed at the well head with a water-tight painted steel standpipe and locking well cover as shown on the monitoring well schematic. A well pad will be constructed so that surface drainage will slope away from the casing.

8. WELL DEVELOPMENT

The monitoring well will be purged and developed with a surge block and/or pump after at least 48 hours and after the annular seal has cured sufficiently to prevent damage during well development. The monitoring well will be purged of at least 10 well volumes. Development purging should continue until the measured depth of the well essentially agrees with the constructed depth, and the parameters of temperature, pH, and electrical conductivity (EC) stabilize, and the water is visually clear of suspended solids. Disposal of development water will be on site or in accordance with local regulations.

9. WELL SAMPLING

Groundwater samples will be collected from each monitoring well at least 48 hours following well development and purging. Depth to water measurements will be taken prior to sampling each well to establish the direction of the hydraulic gradient. Disposable or dedicated sampling equipment may be used to sample the wells. If non-dedicated equipment is used, proper decontamination procedures shall be followed both prior to sampling and between sampling of each well.

After collecting water level measurements, the monitoring wells will be purged of at least three well volumes and sampled. Temperature, pH, and EC will be measured until the parameters stabilize during the purging process. Once the water parameters have stabilized, water samples will be collected in appropriate containers supplied by the laboratory. The water samples may be filtered in the field or by the contracting laboratory. Monitoring well purging and sampling will be conducted in accordance with USEPA SW-846.

10. WELL SURVEY

The monitoring wells will be surveyed for future water gradient determination under the direction of a State registered civil engineer or land surveyor. The well elevations will be measured to the nearest 0.01 foot and referenced to state plane coordinates, or an assumed datum. The survey will be performed in accordance with Geotracker requirements, as applicable.

PROTOCOL FOR GROUNDWATER SAMPLING FOR GENERAL PHYSICAL/GENERAL MINERAL MONITORING

GENERAL SCOPE

The procedures used by Provost and Pritchard Consulting Group (Provost & Pritchard) for purging, sampling and analyses of groundwater samples were developed to obtain representative, consistent, reliable, and reproducible data. The procedures described herein are intended to be of general use. As work progresses, appropriate revisions may be made and approved in writing by Provost & Pritchard's project manager.

Protocols for collecting grab samples from temporary sampling points are similar to protocols from installed monitoring wells. Exceptions are noted in Section 6 below.

PROCEDURES FOR GROUNDWATER SAMPLING

Field work and reporting will be performed under the direction of an appropriately registered or certified professional (Business and Professions Code Sections 6735 and 7835). All field equipment will be calibrated prior to field use, and recalibrated, as needed.

Provost & Pritchard prepared a work plan for the over sight agency review, and obtained appropriate permits, if applicable.

1. EQUIPMENT LIST

The water technician will require the following equipment for purging and sampling:

Decontamination Equipment

- 2 – 5-gallon buckets
- Simple Green or non-phosphate detergent
- small head long handled scrub brush
- 1 ½" or 2" bottle brush

Purging Equipment

- Waterra Powerlift II actuator, Grundfos pump, or disposable bailers
- Horiba multimeter with calibration solutions
- Solinst water level indicator
- DO meter or other as needed
- extra Waterra tubing, footer valves, and surge block if using the Waterra system
- 2 – 5-gallon buckets
- generator with gas
- extension cord
- tie down strap or bungie
- tool bucket (wrenches, pliers, all thread, zip ties)
- latex gloves

Sampling Equipment

- sample bottles
- gallon self-sealing plastic bags
- ice chest with ice
- eye and ear protection
- field camera
- clipboard or forms box with the following:
 - o client contact information
 - o site map with well locations
 - o field purge records
 - o daily field records
 - o waterproof fine point marker and ball point pens
 - o sample labels
 - o chain-of-custody forms

2. EQUIPMENT CLEANING

When dedicated purging and sampling equipment is not used, equipment that may come in contact with the sample will be thoroughly cleaned prior to arrival to the project site. Non-disposable bailers and positive gas-displacement bladder pumps will be disassembled, steam-cleaned, rinsed with (steam-distilled) water, and then reassembled. Wires, hoses and connectors will be cleaned in a similar manner.

3. WATER LEVEL MEASUREMENT

Prior to each sampling event, the static depth to water and depth to the bottom of the well will be measured in each monitoring well. These data will be used to calculate the elevation of the water surface and the required purge volume. The well depth measurement will also be used to monitor siltation.

Generally, a weighted water level meter will be used to measure the depth to groundwater and bottom-of-well measurements. All measurements will be recorded to the nearest 0.01 foot.

4. WELL PURGING

In order to obtain a representative sample of the groundwater contained within the saturated zone, stagnant water within the well casing and filter material must be removed, and fresh formation water allowed to replace it. Removal of the stagnant water is accomplished by pumping or bailing the water contained within the well. Purged water may be dispersed on site or stored in holding tanks or drums for future disposal as applicable.

Where dedicated purging and sampling devices have not been installed, one or more of the following temporary purging devices may be used:

- a positive gas-displacement, TeflonTM and/or stainless steel-housed TeflonTM bladder pump;
- a TeflonTM or stainless steel bailer with bottom discharge unit;

- a stainless steel submersible pump with galvanized piping;
- a peristaltic pump;
- a centrifugal pump;
- a two-stage air-lift pump (Teflon™ or stainless steel); or
- disposable polyethylene bailers with polypropylene check ball.

The purging rate used at a particular monitoring well will depend on the expected or known hydraulic yield.

When purging a low-yield well (one that yields less than three casing volumes prior to being purged to dryness), the well will be purged to dryness twice. When the well recovers the third time, and when it contains a sufficient volume of water for the required analyses, samples will be collected. At no time will a well be purged to dryness if the rate of recharge is such that formation water will cascade down the sides of the casing, or if a purge rate of greater than one-quarter gallon per minute can be maintained.

Groundwater samples will be removed from a monitoring well of moderate- to high-yield only after a minimum of three casing volumes have been purged from the well casing, and purging has been of sufficient duration to result in stabilization of pH, temperature, and electrical conductivity (EC) measurements.

In moderate- to high-yield formations, the purging devices will be placed near the top of the screened interval to ensure that fresh formation water will move upward in the screen. When purging low-yield formations, water will be removed from the bottom of the screened portion. If a Waterra dedicated sampling system is in place, the purging device will be placed in the lower half of the screened portion.

A field sampling log will be maintained for each sampling event and will include the following:

- sampler's identification;
- monitoring well identification;
- climatic conditions;
- depth to water prior to purging;
- type of purging and sampling device;
- purging rate and volume;
- relative well yield volume;
- field parameter measurements (pH, temperature, EC);
- type and number of samples collected; and
- date and time collected.

The pH, temperature, and EC parameters will be monitored and recorded during the purge operations for each casing volume purged. Stabilization of pH, temperature, and EC parameters will be indicated by values within 10% of one another for a minimum of three consecutive readings. Field parameters will be measured using a pH meter calibrated to standard buffers, and an electrical conductivity meter equipped with a thermometer. Field

equipment will be standardized at the beginning of each use, according to the manufacturers' specifications and consistent with SW-846.

5. SAMPLE COLLECTION

Samples of fresh formation water will be collected only after the appropriate volume of water has been purged from the casing, and field parameters have stabilized.

To increase the likelihood that groundwater samples are representative of the groundwater contained within the formation, it is important to minimize physical or chemical alteration of the sample during the collection process. If dedicated sampling devices are not used, the following procedures will be strictly adhered to:

- Sampling equipment will be thoroughly cleaned in accordance with Section 2 (Equipment Cleaning) of this protocol.
- Blank samples, if required, will be collected during the final rinse of the sampling equipment cleaning process.

A general order of collection for some common groundwater analyses follows:

- total organic carbon
- extractable organic compounds
- total metals
- dissolved metals
- phenols
- sulfate and chloride
- turbidity
- nitrate and ammonia

6. TEMPORARY SAMPLING POINT EXCEPTIONS

Once the borehole is terminated, a temporary well screen will be placed at the borehole bottom. Sand will be emplaced in the annular space surrounding the temporary screen. Groundwater will be purged by disposable bailer following standard protocol, as possible. Grab sampling will be performed by disposable bailer. Sample handling will follow standard protocol. Unless otherwise specified in the workplan, the temporary sampling point will be removed and the borehole filled in accordance with appropriate local guidelines and our protocols for soil borings.

7. SAMPLE PACKAGING AND TRANSPORT

Sample Preservation

Many chemical analytes and physical parameters monitored in groundwater are not chemically stable, and therefore require chemical preservation. Sample preservation will be in accordance with current standards. Sample containers and preservatives will be supplied by the contract laboratory. Samples will be held in ice in a covered, secured ice chest specifically designated for the purpose of sample storage and transport.

Sample Filtration

For some analyses, samples may require filtration to remove suspended particles. As needed, Provost & Pritchard will filter samples in the field. Unless otherwise specified, 45-micron cellulose acetate filters will be used. One of two filter types will be used: an in-line filter which is used in conjunction with a sample pump, or a pressure or vacuum cylinder filter which will be used in conjunction with a bailer.

Sample Labeling

Sample containers will be labeled in the field. Labels will contain the following information:

- consultant's identification;
- project number or identification;
- sampler's identification;
- date and time of collection; and
- sample identification.

Sample Transport

All samples will be delivered to the laboratory within a time frame to allow for analyses within the appropriate holding time. Sealed sample containers will not be opened by other than the laboratory personnel who will perform the requested analyses.

Custody Seal

If it is necessary for samples or sample chests to leave Provost & Pritchard's control prior to delivery to the laboratory a custody seal will be placed on each sample container and/or sample chest to discourage tampering during transportation. The custody seal will contain the sampler's signature, and the date and time the seal was emplaced.

Chain of Custody

In order to document and trace sample possession, a positive signature chain-of-custody record will accompany the sample through the laboratory analyses. The completed chain-of-custody record will be included in the laboratory's final report.

8. SAMPLE ANALYSES

Groundwater samples will be analyzed by a California Certified Environmental Laboratory Accreditation Program (ELAP) laboratory.

Requests for sample analyses will be made in writing and will be included as part of the chain-of-custody record.

FIELD QUALITY ASSURANCE/QUALITY CONTROL

Travel, equipment, and field blanks will be collected as appropriate and handled and transported in the same manner as the groundwater samples. Equipment blanks will be collected at a rate

of one per sampling episode by circulating steam-distilled water through cleaned sampling equipment during the final rinse stage.

LABORATORY QUALITY ASSURANCE/QUALITY CONTROL

Duplicate samples may be collected at a rate of one duplicate per ten samples, or one duplicate per sampling episode, if fewer than ten samples are collected. Duplicate samples will be delivered to the primary laboratory to evaluate the laboratory's reliability.

Depending on the project sampling and analyses plan or at the discretion of the project manager, split samples may be collected. The split sample will be handled the same as the primary sample, but will be delivered to a second laboratory. A comparison of the split sample results will be made to further evaluate the primary laboratory's performance.

Duplicate and/or split samples collected from a single well will be collected from a single casing volume when possible. When a single casing volume is insufficient, samples will be collected in as rapid a succession as possible.

Quality assurance/quality control sample analytical data will be used to monitor the laboratory performance, sampling technique, and as indicators of potential sample analyses or sample collection anomaly.

For general minerals analyses, a cation/anion balance will be calculated by the laboratory as an error check.

Appendix D-Health and Safety Plan

HEALTH AND SAFETY PLAN

Riverdale PUD – Riverdale, Fresno County, California – Monitoring Well Installation Work Plan

INTRODUCTION

This Health and Safety Plan (**HSP**) was prepared by Provost & Pritchard Consulting Group (**Provost & Pritchard**) for the subject site. This HSP complies with, but does not replace, Federal Health and Safety regulations as set forth in 29 CFR 1910 and 1926, CA Health and Safety Regulations as set forth in Title 8, CA Code of Regulations, and guidance established by the CA Department of Health Services. All on-site personnel are to use this plan as a supplement to such rules, regulations, and guidelines.

If the Groundwater Monitoring Well Construction activities are contracted through Provost & Pritchard, this Health and Safety Plan will be applicable.

RESPONSIBILITIES

Provost & Pritchard will have site health and safety oversight and coordinate responsibilities for Provost & Pritchard employees. A copy of this HSP will be kept on-site. All on-site Provost & Pritchard employees must read and sign the Provost & Pritchard Health and Safety Compliance Form.

Project Manager: Maija Madec, phone (559) 326-1100

The Project Manager (**PM**) directs on-site sampling, investigations and operational efforts of Provost & Pritchard personnel. The PM may delegate all or part of these duties to an appointed Site Safety Officer (**SSO**).

Site Safety Officer: Shawn Vaughn, phone (559) 326-1100

The Site Safety Officer (**SSO**)'s duty is to protect the health and safety of Provost & Pritchard personnel and may be carried out by the PM or other site manager.

Regional Health and Safety Manager: Kim Hansen 559-449-2700

The Regional Health and Safety Manager (**RHSM**) periodically audits operations for compliance with this plan, and provides health and safety support as requested by the SSO and PM.

SCOPE OF WORK

Borings to a maximum depth of approximately 260 feet will be drilled by qualified contractors to collect soil samples and install up to three monitoring wells. Equipment is listed in Table 1.

Table 1. Protective and Safety Equipment List

R	Hard hat	R	Safety glasses with side shields
R	Chemical resistant steel toed boots	AN	Ear plugs or muffs
R	Work gloves	AN	Traffic control (cones, caution tape, etc.)
NR	Nitrile® gloves	NR	Wash tubs, scrub brushes, and detergent
R	Long pants	R	First aid kit
AN	Shirt with sleeves	AN	Drinking water
NR	PID/FID		

R – Required

AN – As needed

NR – Not required

IDENTIFIED HAZARDS

Physical hazards that may be encountered on site include those from drilling and sampling operations, heat stress, underground and overhead utilities, on-site traffic, and equipment failures. These are discussed below.

Heat Stress Recognition and Control

Evidence for heat stress includes flush skin, rash, irritability, faintness, and dry, cool skin. Heat stress is caused by a number of interacting factors, including environmental conditions, clothing, work load, and individual worker characteristics. Rest periods, adequate drinking water intake, and regular monitoring for signs and symptoms of distress are vital.

To help prevent heat stress, site personnel should drink a sufficient quantity of water to maintain body fluids at normal levels. The normal thirst mechanism is not sensitive enough to provide for sufficient consumption of water to replace lost fluids. When heavy perspiration occurs 1) maintain drinking water temperature at 50°F to 60°F, 2) drink 16 ounces of fluid (preferably water or electrolyte replenishing fluid, such as Gatorade) before beginning work, and 3) drink a cup or more of water or equivalent fluid every 15 to 20 minutes, or at each monitoring break. A total of 1 to 1.6 gallons of fluid per day is recommended, but more may be necessary to maintain body weight.

Underground Utilities

The driller will attempt to locate underground utility locations prior to commencement of drilling and sampling activities. Resources may include site plans, utility companies, Underground Service Alert (**USA**) (800-642-2444), and hand auguring methods.

Traffic Hazards

Orange traffic cones and/or barricades may be used to delineate an exclusion zone around the work site. Barriers should be set at a 10-foot radius (as practical) around the work area. Unauthorized visitors will not be allowed in the work/exclusion zone.

SAFE WORK PRACTICES

Eating, drinking, chewing gum or tobacco, and smoking are prohibited in areas that are contaminated or potentially contaminated, or where the possibility for the transfer of contamination exists. Avoid personal contact with potentially contaminated substances and wet areas. Avoid kneeling, leaning, or sitting on equipment or ground.

EMERGENCY RESPONSE PLAN

Emergency telephone numbers are presented at the end of this HSP. A cellular telephone is required on site. A first aid kit is kept available at the site for use in case of minor injuries. If direct contact with contaminants occurs, affected skin areas are immediately washed with soap and water.

Incident Report

In the event of an injury or illness, work is to be stopped until the SSO and the RHSM have determined the cause and have taken appropriate action. Any injury or illness is to be reported on the Provost & Pritchard Accident, Injury and Illness Investigation form.

Operations Shutdown

Under extreme hazardous situations, the PM, SSO, or subcontractor's Safety Representative will request that sampling procedures be temporarily suspended while the underlying hazard is corrected or controlled. The SSO has ultimate authority for shutdown and restart of operations.

EMERGENCY TELEPHONE NUMBERS

SITE CONTACT: Vincent Romero (559) 867-3838

TABLE 1: Telephone Numbers for Fire Department, Police, and Ambulance.

<u>Fire Department/Emergency Response/Police/Ambulance</u>	<u>Telephone</u>
Fire Station	911
Sheriff's Department/Highway Patrol	911
Ambulance	911

TABLE 2: Nearest Hospital or Care Facility

<u>City</u>	<u>Hospital or Facility</u>	<u>Address</u>	<u>Telephone</u>
Hanford	Adventist Health Hanford	115 Mall Dr, Hanford, CA 93230	(559) 582-9000

Directions to Hospital from Site:

1. Head south on S Brawley Avenue 0.5 Miles
2. Turn left onto W Harlan Ave 1.0 Miles
3. Turn right onto S Marks Ave 3.0 Miles
4. Turn left onto W Excelsior Ave 9.9 Miles
5. Turn right onto 12th Ave 4.7 Miles
6. Turn left onto Kings County Dr 0.5 Miles
7. Continue onto Mall Dr 0.3 Miles
8. The hospital will be on the left

HEALTH AND SAFETY COMPLIANCE

I, the undersigned, have received a copy of the Health and Safety Plan for the project identified below. I have read the plan, understand it, and agree to comply with all of the health and safety requirements therein. I understand that I may be prohibited from continuing work on the project for failing to comply.

I have been briefed by a project safety authority on the health and safety requirements of the project.

PROJECT NUMBER _____ **Date of Plan** _____

PROJECT TITLE _____

FIRM _____

Print Name	Signature	Date
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Print Name	Signature	Date
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Print Name	Signature	Date
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Print Name	Signature	Date
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Print Name	Signature	Date
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Print Name	Signature	Date
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Print Name	Signature	Date
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ACCIDENT INVESTIGATION REPORT

Accident investigated by: _____ Date: _____ Time: _____

Injured Person's Name: _____

Date of Accident: _____ Time of Accident: _____ Date Accident Reported: _____

Where did accident occur? _____

Injured Person's Statement on Details of Accident _____

Names of witness(es): _____

Witness Statement: _____

Investigators comments/description of accident: _____

Photographs attached _____yes _____no (If no, explain) _____

Reason for Accident:

- ☐ Lack of knowledge/Experience
- ☐ Not paying attention
- ☐ Playing around
- ☐ Improper Attire
- ☐ Repair needed
- ☐ Other _____

Recommendations to avoid similar accidents: _____

Investigator's Signature: _____ Date _____

HAZARD ASSESSMENT AND CORRECTION RECORD

Person Conducting Inspection:

Unsafe Condition or Work Practice:

Corrective Action Taken: